

GSM8K · \$5.1

Qwen2.5-14B-Instruct · lm-evaluation-harness 0.4.11 · 5-shot

Plain few-shot (direct-answer)

```
[sys] You are Qwen, created by Alibaba Cloud.

Q: Jen and Tyler are gymnasts ...

A: ... #### 12

Q: Four people in a law firm ...

A: ... #### 1

... [3 more few-shot examples] ...

Q: Janet's ducks lay 16 eggs/day.

    She eats 3 for breakfast; bakes 4 into muffins.

    She sells the rest at $2 each. How much?

A: 16-3-4 = 9 eggs.  9 × 2=18.  #### 18
```

↓ Δ = -56.0 pp

Chain-of-thought few-shot

```
[sys] You are an expert problem solver.

    Think step by step, then state the answer.

Q: Jen and Tyler are gymnasts ...

A: ... #### 12

... [same 5 few-shot examples] ...

Q: Janet's ducks lay 16 eggs/day. ...

A: Janet uses 3 + 4 = 7 eggs each day.

    16 - 7 = 9 eggs remain.

    9 × 2=18 per day.

The answer is $18 per day.
```

Same model · same harness · same 5-shot context.
CoT prompt elicits step-by-step prose;
the ##### delimiter regex finds nothing.
Parser records 0 for correctly-answered items.

Prompts: run logs · experiments/controlled_eval/5-shot-GSM8K/
Scores verified: 58.5% plain, 2.5% CoT (limit=200, seed=42/123/7)

HellaSwag · \$5.2

Gemma-7B · two papers, two protocols

Phi-3 paper · 5-shot · generation-based

```
# 5-shot generation template

# (canonical; verbatim not published)

Context: She bopped along, but the music

    was really loud, so she turned it down.

    Now, how does the next sentence go?

A) She turned it up more.

B) She left the store.

C) She decreased the volume.

D) She plugged her ears.

Answer: C ← free-text generation
```

↑ Δ = +31.4 pp

Gemma paper · 0-shot · log-likelihood

```
# 0-shot log-likelihood template

# (canonical; verbatim not published)

Context: She bopped along, but the music

    was really loud, so she turned it down.

Completion A: She turned it up more.

Completion B: She left the store.

Completion C: She decreased the volume.

Completion D: She plugged her ears.

← rank by log P(completion | context)
```

Same benchmark label · two dimensions differ.
Scoring mode: generation vs. log-likelihood.
N-shot: 5-shot vs. 0-shot.
Primary driver: scoring mode, not n-shot.

Scores: arXiv:2404.14219 Table 3 (Phi-3, 49.80)
arXiv:2403.08295 Table 6 (Gemma, 81.20)

BBH · Gemma-2-9B · \$5.3

Five sources · two score clusters · two protocols

OLv2 & SEA-LION · lighteval · 3-shot · log-likelihood

```
# lighteval BBH protocol (OLv2 / SEA-LION)

Q: Which statement is logically false?

    given: all mammals breathe air;

    whales are mammals.

(A) Whales breathe air.  (B) Whales swim.

(C) Whales are reptiles. (D) Whales have lungs.

rank: argmax log P(choice | context)

← 3-shot; log-likelihood ranking

OLv2 = 34.10  exact 2-decimal match

SEA-LION = 34.10  ↳ shared pipeline origin
```

↑ Δ = +34.8 pp

Gemma 2 / Gemma 3 reports · CoT generation

```
# CoT generation protocol (Suzgun et al. 2023)

Q: Which statement is logically false? ...

Let's think step by step.

All mammals breathe air; whales are mammals

therefore whales breathe air. That is true.

Whales are not reptiles – that is false.

The answer is (C).

Gemma 2 = 68.20  Gemma 3 = 69.00

Gemma 3 = 69.40  (independent)
```

Two score clusters — two protocols.
OLv2 & SEA-LION: exact 2-decimal match;
shared pipeline, not independent evidence.
Gemma 2/Gemma 3 reports: CoT generation.

Scores: data/aggregated/all_results.csv
arXiv:2504.05747 (SEA-LION, Table 3)
BBH cluster 2 mean: (68.20+69.00+69.40)/3 = 68.87